fn log_inv_gamma_hi_from_x

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Proves soundness of `fn log_inv_gamma_hi_from_x` in `mod.rs`.

1 Hoare Triple

Precondition

`x` is finite.

Pseudocode

```
1 def log_inv_gamma_hi_from_x(x: float):
2     gamma_lo = gamma_lo_from_x(x)  #  propagates helper errors
3     ensure_open_unit_interval(gamma_lo)  #  propagates helper errors
4     gamma_minus_one_lo = gamma_lo.neg_inf_sub(1.0)  #  propagates helper errors
5     ln_gamma_lo = gamma_minus_one_lo.neg_inf_ln_1p()  #  propagates helper errors
6     return -ln_gamma_lo
```

Postcondition

Theorem 1.1. For any finite `x`, `log_inv_gamma_hi_from_x` either returns `Err(e)`, or returns `Ok(out)` where

$$\texttt{out} \geq \log\left(\frac{1}{1 - e^{-\texttt{x}}}\right).$$

2 Proof

Assume the precondition and write $\gamma = 1 - e^{-x}$.

Lemma 2.1. `log_inv_gamma_hi_from_x` only returns `Err(e)` if one of the fallible calls on lines 2, 3, 4, or 5 returns `Err(e)`.

Proof. These are exactly the fallible operations in the pseudocode; the final negation is infallible. □

Lemma 2.2. If `log_inv_gamma_hi_from_x` returns `Ok(out)`, then

$$\texttt{out} \geq \log\left(\frac{1}{1 - e^{-\texttt{x}}}\right).$$

Proof. If line 3 succeeds, then `gamma_lo` lies in $(0, 1)$. By the proof of `gamma_lo_from_x`,

$$\texttt{gamma_lo} \leq \gamma.$$

Subtracting 1 preserves order, so

$$\texttt{gamma_minus_one_lo} \leq \gamma - 1.$$

Both values lie in $(-1, 0)$, so `neg_inf_ln_1p` is applied on a valid domain and yields

$$\texttt{ln_gamma_lo} \leq \ln(\gamma).$$

Negating reverses the inequality:

$$\texttt{out} = -\texttt{ln_gamma_lo} \geq -\ln(\gamma) = \log\left(\frac{1}{1 - e^{-x}}\right).$$

□

Proof of Theorem 1.1. Holds by Lemma 2.1 and Lemma 2.2. □